Juniper Networks® vMX Release Notes

Release 16.2R1
28 November 2016

These release notes accompany this release of the Juniper Networks® virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

Contents

Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We recommend Ubuntu as the host OS.

vMX allows you to leverage Junos OS Release 16.2 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

This release of vMX supports most of the features available on Juniper Networks MX Series routers with the following exception:

- High availability features such as virtual Routing Engine redundancy are not supported in this release.

New and Changed Features

This section describes the new features and enhancements in this release.

- **Class of Service support for eight queues**—Two-level hierarchical scheduling (per-unit scheduler or hierarchical scheduler) with VLAN queuing is supported for eight queues. Each VLAN uses three traffic classes and eight queues. There are two high-priority queues (Queue 0 and Queue 6), two medium-priority queues (Queue 1 and Queue 7), and four low-priority queues (Queue 2 through Queue 5). Shaping is supported at the traffic class level, not at the queue level. Weighted random early detection, drop profiles, and queue buffer size configuration are not supported. You enable CoS by configuring the **flexible-queuing-mode** option at the [**edit chassis fpc 0**] hierarchy level.
- **ESXi hypervisor support**—VMware ESXi 5.5 supports SR-IOV as physical PCI device. You can install vMX using an OVA image.
- **Licenses**—Evaluation licenses are perpetual. You can download the vMX software and use the BASE application package with 1 Mbps bandwidth without a license indefinitely. In previous releases, you had only 30 days to evaluate the software without a valid license.

  If you upgrade from a BASE package license to an ADVANCE or PREMIUM package license or if you downgrade from an ADVANCE or PREMIUM package license to a BASE package license, you must restart the routing protocol process. If your configuration has logical systems, you must restart the routing protocol process for all logical systems.
- **Performance mode is default for chassis**—Performance mode is enabled by default for the chassis. Performance mode needs more vCPUs and memory to run at higher bandwidth, while lite mode needs fewer vCPUs and memory to run at lower bandwidth. Make sure you have configured the proper number of vCPUs and memory for your VMs. We recommend you enable hyperthreading in BIOS.

To tune performance mode for unicast traffic, you can change the number of Workers dedicated to processing multicast and control traffic. The default specifies that all available Workers are used to process all traffic. You specify the number of dedicated Workers by configuring the **number-of-ucode-workers** ***number-workers*** option at the [**edit chassis fpc 0 performance-mode**] hierarchy level.

- **Support for 96 ports using lite mode**—You can use up to 96 ports for virtio running in lite mode. Other configurations running in performance mode support up to 23 ports.
- **Support for FreeBSD 10 kernel for Junos OS**—FreeBSD 10 is the underlying OS for Junos OS instead of FreeBSD 6.1 for vMX. With FreeBSD 10, you can allocate multiple vCPUs to the VCP.
- **Support for X710 NICs**—You can use Intel X710 PCI-Express NICs. Before you install vMX, make sure you have updated the drivers.
- **Virtual broadband network gateway**—Using vMX, you can deploy a virtual broadband network gateway (vBNG) on x86 servers to terminate broadband subscribers. vBNG supports most of the subscriber management features available with Junos OS Release 16.2 on MX Series routers.

  vBNG runs on vMX, so it has similar exceptions; the following subscriber management features available on MX Series routers are not supported for vBNG:

  - High availability features such as hot-standby backup for enhanced subscriber management and MX Series Virtual Chassis.
  - CoS features such as shaping applied to an agent circuit identifier (ACI) interface set and its members.

  To deploy a vBNG instance, you must purchase these licenses:

  - vMX PREMIUM application package license with 1 Gbps, 5 Gbps, 10 Gbps, or 40 Gbps bandwidth
  - vBNG subscriber scale license with 1000, 10 thousand, 100 thousand, or 1 million subscriber sessions for one of these tiers:

Tier	Description
Introductory	L2TP features including L2TP LNS services, secure policy, service activation and deactivation
Preferred	Features in the Introductory tier, and DHCP subscriber services, PPP/LAC subscriber services, DHCP relay and DHCP local server
Elite	Features in the Preferred tier, and pseudowire ingress termination

With the appropriate vMX PREMIUM license, you can evaluate vBNG without a vBNG subscriber scale license for 30 days. After 30 days, you are limited to 10 subscriber sessions.

[See Junos OS Broadband Subscriber Management and Services Library.]

Known Behavior

This section contains the known behaviors and limitations in this release.

- Scale limitation is observed with VLAN tag operation and circuit cross-connect (CCC).
- When vMX is deployed, the management port is not configured, so you must use the serial console for configuration. Only a small number of configuration lines can be pasted in the vMX console. As a workaround, perform initial configuration to set the root password and to allow SSH access in the vMX console and perform further configuration using SSH.
- ICMP echo request packets are handled inline, which means ICMP packets do not reach the VCP and are replied to by the VFP itself. No ICMP packets are seen on the VCP and packet capture tools do not capture ICMP packets on the VCP.

Known Issues

This section lists the known issues in this release.

- When the FPC is restarted, **kernel: GENCFG: op 32 (Resync blob) failed** syslog message appears. This harmless message can be ignored. PR1050467
- LACP packets are getting dropped on the bridge. PR1059231
- LLDP packets are getting dropped on the bridge (for virtio). PR1066850
- When the FPC is restarting, **COS(cos_chassis_scheduler_pre_add_action:2137)** error messages appear indicating that flexible queuing mode is not enabled. PR1070655
- Output for the **show pfe statistics traffic** command might take a few seconds to appear and many **MIB2D_COUNTER_DECREASING: pfes_stats_delta: counter** messages appear in the log file before the response time improves. PR1071659
- When the FPC is restarting, a **Received unsupported pic_mask 0x1 ignored message** message appears in the syslog file. PR1072436
- Traffic loss occurs at a remote receiver because of lost remote PIM joins to the local receiver. PR1087031
- The VMXNET3 driver does not support large MTU. PR1120215
- When committing the configuration, **%DAEMON-4-SNMP_EVLIB_FAILURE: PFED ran out of transfer credits with PFE.Failed to get stats.** messages sometimes appear in the syslog file. PR1124902
- Multiple vMX instances using SR-IOV on the same host are not supported. PR1130534
- When committing the configuration, **pfed: %USER-3: downward spike received from pfe for opackets_reply:3545 opackets_record:48932521** messages sometimes appear in the syslog file. PR1146002
- Many **clksync_update_tod_to_pfe:109 Failed to send non-PEC pfe TOD update to other PFEs. Error code: 29.** messages appear in the syslog file. PR1148344

- Scheduler slips are occurring in large VPLS configurations and displaying **JTASK_SCHED_SLIP** log messages. PR1153290
- Observing **rpd[7116]: %DAEMON-6: ifl_delete: ifl** error messages that indicate a multicast tunnel (mt) has been deleted. PR1156725
- CPU pinning is not working properly with OVS bridges. PR1171096

Subscriber Management

- Dual-stack (DHCPv6 over PPPoE) subscribers can be lost on an LNS after the subscriber management process is restarted on vMX. PR1165378
- When bringing up an LNS session on vMX, the Cos-Shaping-Rate attribute (ERX-Attr-177) is sometimes omitted from the Acct-Start messages sent to RADIUS. PR1167154
- After adding or deleting the l2tp-inline-lns license on vMX and rebooting vMX, fxp0.0 was not reachable. As a workaround, access vMX through the console and reboot vMX. PR1173990

System Requirements

These topics provide system requirements for each supported environment.

Minimum Hardware and Software Requirements for KVM

Table 1 on page 5 lists the hardware requirements.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Sample system configuration	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type, use Intel 82599-based PCI-Express cards (10 Gbps) and Ivy Bridge processors.
Number of cores NOTE: Performance mode is the default mode and the minimum value is based on one port. To calculate the minimum number of vCPUs needed by VFP for performance mode: (4 * *number-of-ports*) + 3	For lab simulation use case applications (lite mode): Minimum of 4 • 1 for VCP • 3 for VFP For low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications (performance mode): Minimum of 8 • 1 for VCP • 7 for VFP

Table 1: Minimum Hardware Requirements for KVM (continued)

Description	Value
Memory NOTE: Performance mode is the default mode.	For lab simulation use case applications (lite mode): Minimum of 5 GB • 1 GB for VCP • 4 GB for VFP For low-bandwidth or high-bandwidth applications (performance mode): Minimum of 16 GB • 4 GB for VCP • 12 GB for VFP Additional 2 GB recommended for host OS
Storage	Local or NAS
Other requirements	Intel VT-d capability Hyperthreading AES-NI

Table 2 on page 6 lists the software requirements for Ubuntu.

Table 2: Minimum Software Requirements for Ubuntu

Description	Value
Operating system	Ubuntu 14.04 LTS (recommended host OS) Linux 3.13.0-32-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11 or later
Required packages NOTE: Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev NOTE: libvirt 1.2.19

Minimum Hardware and Software Requirements for VMware

Table 3 on page 7 lists the hardware requirements.

Table 3: Minimum Hardware Requirements for VMware

Description	Value
Number of cores NOTE: Performance mode is the default mode and the minimum value is based on one port. To calculate the minimum number of vCPUs needed by VFP for performance mode: (4 * *number-of-ports*) + 3	For performance mode: Minimum of 8 • 1 for VCP • 7 for VFP For lite mode: Minimum of 4 • 1 for VCP • 3 for VFP
Memory NOTE: Performance mode is the default mode.	For performance mode: Minimum of 16 GB • 4 GB for VCP • 12 GB for VFP For lite mode: Minimum of 10 • 2 GB for VCP • 8 GB for VFP
Storage	Local or NAS

Table 4 on page 7 lists the software requirements.

Table 4: Software Requirements for VMware

Description	Value
Hypervisor	ESXi 5.5 Update 2
Management Client	vSphere 5.5 or vCenter Server

Verifying Security Signatures

The vMX image is securely signed, so you can verify the signature for the image.

To verify the signature:

1. Download the public key certificate and the software image from the Juniper Networks Download Software page.

2. Import the public key into a temporary GPG key ring.

   ```
   mkdir temp
   cd temp
   gpg --homedir . --import certificate-file
   ```

   For example:

gpg --homedir . --import Juniper_vMX_public_key_2015.asc

```
gpg: WARNING: unsafe permissions on homedir `.'
Warning: using insecure memory!
gpg: keyring `./secring.gpg' created
gpg: keyring `./pubring.gpg' created
gpg: ./trustdb.gpg: trustdb created
gpg: key CA6E E4DD E89A EB4C F22F  6897 7B82 9893 BA75 0B9B: public key "vMX
Trusted Kernel 2015 <ca@juniper.net>" imported
gpg: Total number processed: 1
gpg:               imported: 1  (RSA: 1)
```

NOTE: The hexadecimal key value and name vary with each key. The hexadecimal value is an easy way to confirm that the key is authentic.

3. Verify the validity of the signature.

```
gpg --homedir . --verify signature-file file-to-be-signed
```

For example:

gpg --homedir . --verify vmx-14.1R5.4-1.tgz.sig jinstall-vmx-14.1R5.4-domestic-signed.tgz

```
gpg: WARNING: unsafe permissions on homedir `.'
Warning: using insecure memory!
gpg: Signature made Tue Jul  7 16:50:05 2015 PDT using RSA key ID BA750B9B
gpg: Good signature from "vMX Trusted Kernel 2015 <ca@juniper.net>"
gpg: This key is certified with a trusted signature!
gpg: WARNING: This key is not certified with a trusted signature!
gpg:          There is no indication that the signature belongs to the owner.
Primary key fingerprint: CA6E E4DD E89A EB4C F22F  6897 7B82 9893 BA75 0B9B
```

The second warning message appears because the key has not been marked as trusted. You can ignore the warning messages.

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or JNASC support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf .
- Product warranties—For product warranty information, visit https://www.juniper.net/support/warranty/ .
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: https://www.juniper.net/customers/support/
- Search for known bugs: http://www2.juniper.net/kb/
- Find product documentation: https://www.juniper.net/documentation/
- Find solutions and answer questions using our Knowledge Base: https://kb.juniper.net/
- Download the latest versions of software and review release notes: https://www.juniper.net/customers/csc/software/
- Search technical bulletins for relevant hardware and software notifications: https://www.juniper.net/alerts/
- Join and participate in the Juniper Networks Community Forum: https://www.juniper.net/company/communities/
- Open a case online in the CSC Case Management tool: https://www.juniper.net/cm/

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: https://tools.juniper.net/SerialNumberEntitlementSearch/

Opening a Case with JTAC

You can open a case with JTAC on the Web or by telephone.

- Use the Case Management tool in the CSC at https://www.juniper.net/cm/ .
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see https://www.juniper.net/support/requesting-support.html .

Revision History

28 November 2016—Revision 1, Junos OS Release 16.2R1—vMX